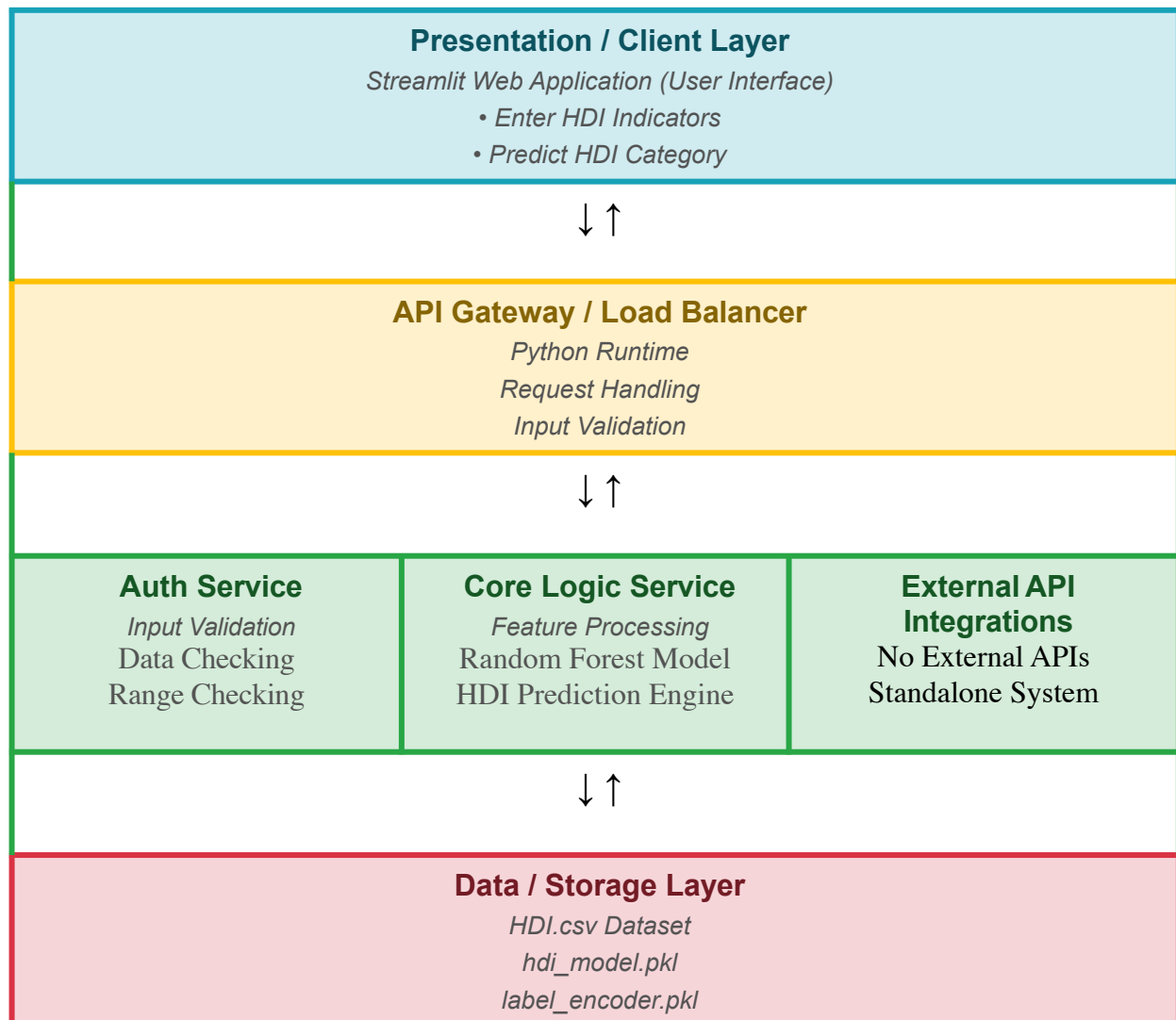


Solution Architecture

Date	2 July 2026
Team ID	
Project Name	Human Development Index (HDI) Predictor – A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides an interactive web interface where users enter Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita, then displays the predicted HDI category.	Streamlit , HTML, CSS
API Gateway	Receives user input, validates the request, and forwards it to the prediction engine. It also returns the prediction result to the user interface.	Streamlit Backend, Python
Microservices	Performs input validation and verifies that the entered values are valid before prediction. (<i>No user login/authentication is implemented.</i>)	Python , Scikit-learn, Pandas, NumPy, Pickle
Database	Stores the HDI dataset used for model training and the serialized trained model used during prediction.	CSV Dataset (<i>HDI.csv</i>), <i>hdi_model.pkl</i> , <i>label_encoder.pkl</i>